



Title

rdmc — Point Estimation and Robust Bias-Corrected Inference for Multi-Cutoff Designs.

Syntax

```
rdmc depvar runvar [if] [in], cvar(string) [ fuzzy(string) derivvar(string)
pooled_opt(string) verbose pvar(string) qvar(string) hvar(string)
hrightvar(string) bvar(string) brightvar(string) rhovar(string)
covsvar(string) covs_dropvar(string) kernelvar(string) weightsvar(string)
bwselectvar(string) scaleparvar(string) scaleregulvar(string)
masspointsvar(string) bwcheckvar(string) bwrestrictvar(string)
stdvarsvar(string) vcevar(string) level(#) all detail vleverage nochecks
nowarnings plot graph_opt(string) conventional ]
```

Description

rdmc implements point estimation and robust bias-corrected inference for regression discontinuity (RD) designs with multiple cutoffs. For methodological background see [Keele and Titiunik \(2015\)](#), [Cattaneo, Keele, Titiunik and Vazquez-Bare \(2016\)](#), and [Cattaneo, Keele, Titiunik and Vazquez-Bare \(2021\)](#). It also computes alternative estimation and inference procedures available in the literature.

Companion commands are: [rdmcplot](#) for data-driven RD plots for multi-cutoff designs, and [rdms](#) for point estimation and robust bias-corrected inference for multi-score designs.

A detailed introduction to this command is given in [Cattaneo, Titiunik and Vazquez-Bare \(2020\)](#).

Companion R functions are also available [here](#).

This command employs the Stata (and R) package [rdrobust](#) for underlying calculations. See [Calonico, Cattaneo and Titiunik \(2014\)](#), [Calonico, Cattaneo and Titiunik \(2015\)](#), and [Calonico, Cattaneo, Farrell and Titiunik \(2017\)](#) for more details.

Related Stata and R packages useful for inference in RD designs are described in the following website:

<https://rdpackages.github.io/>

Options

Estimand

cvar(*string*) specifies the numeric variable containing the RD cutoff for *indepvar* for each unit in the sample.

fuzzy(*string*) indicates a fuzzy design. Use **fuzzy**(*treatvar sharpbw*) to request sharp-RD bandwidth selection for fuzzy RD. See [rdrobust](#) for details.

derivvar(*string*) a variable of length equal to the number of different cutoffs that specifies the order of the derivative for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

rdrobust Options

pooled_opt(*string*) specifies the options to be passed to **rdrobust** to calculate pooled estimates. See [rdrobust](#) for details.

verbose displays the output from **rdrobust** for estimating the pooled estimand.

Local Polynomial Regression

pvar(string) a variable of length equal to the number of different cutoffs that specifies the order of the polynomials for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

qvar(string) a variable of length equal to the number of different cutoffs that specifies the order of the polynomials for bias estimation for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

hvar(string) a variable of length equal to the number of different cutoffs that specifies the bandwidths for **rdrobust** to calculate cutoff-specific estimates. When **hrightvar** is specified, **hvar** indicates the bandwidth to the left of the cutoff. When **hrightvar** is not specified, the same bandwidths are used at each side. See [rdrobust](#) for details.

hrightvar(string) a variable of length equal to the number of different cutoffs that specifies the bandwidths to the right of the cutoff for **rdrobust** to calculate cutoff-specific estimates. When **hrightvar** is not specified, the bandwidths in **hvar** are used at each side. See [rdrobust](#) for details.

bvar(string) a variable of length equal to the number of different cutoffs that specifies the bandwidths for bias estimation for **rdrobust** to calculate cutoff-specific estimates. When **brightvar** is specified, **bvar** indicates the bandwidth to the left of the cutoff. When **brightvar** is not specified, the same bandwidths are used at each side. See [rdrobust](#) for details.

brightvar(string) a variable of length equal to the number of different cutoffs that specifies the bandwidths for bias estimation to the right of the cutoff for **rdrobust** to calculate cutoff-specific estimates. When **brightvar** is not specified, the bandwidths in **bvar** are used at each side. See [rdrobust](#) for details.

rho var(string) a variable of length equal to the number of different cutoffs that specifies the value of rho for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

covs var(string) a variable of length equal to the number of different cutoffs that specifies the covariates for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

covs_drop var(string) a variable of length equal to the number of different cutoffs that specifies whether collinear covariates should be dropped. See [rdrobust](#) for details.

kernel var(string) a variable of length equal to the number of different cutoffs that specifies the kernels for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

weights var(string) a variable of length equal to the number of different cutoffs that specifies the weights for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

Bandwidth Selection

bwselect var(string) a variable of length equal to the number of different cutoffs that specifies the bandwidth selection method for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

scalepar var(string) a variable of length equal to the number of different cutoffs that specifies the value of scalepar for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

scaleregul var(string) a variable of length equal to the number of different cutoffs that specifies the value of scaleregul for **rdrobust** to calculate cutoff-specific estimates. See [rdrobust](#) for details.

masspointsvar(*string*) a variable of length equal to the number of different cutoffs that specifies how to handle repeated values in the running variable. See [rdrobust](#) for details.

bwcheckvar(*string*) a variable of length equal to the number of different cutoffs that specifies the value of **bwcheck**. See [rdrobust](#) for details.

bwrestrictvar(*string*) a variable of length equal to the number of different cutoffs that specifies whether computed bandwidths are restricted to the range of *runvar*. See [rdrobust](#) for details.

stdvarsvar(*string*) a variable of length equal to the number of different cutoffs that specifies whether *depvar* and *runvar* are standardized. See [rdrobust](#) for details.

Variance-Covariance Estimation and Inference

vcevar(*string*) a variable of length equal to the number of different cutoffs that specifies the variance-covariance matrix estimation method for **rdrobust** to calculate cutoff-specific estimates. Current options include **nn**, **hc0**, **hc1**, **hc2**, **hc3**, **cr1**, **cr2**, and **cr3**; cluster-robust options include the cluster variable in the same string, for example **cr2 schoolid**. See [rdrobust](#) for details.

level(#) specifies the confidence level for confidence intervals. See [rdrobust](#) for details.

all, **detail**, and **vleverage** pass the corresponding display and diagnostic options to **rdrobust**.

nochecks and **nowarnings** pass the corresponding validation and warning controls to **rdrobust**.

conventional displays conventional, instead of robust-bias corrected, p-values and confidence intervals.

Plot

plot plots the pooled and cutoff-specific estimates and the weights given by the pooled estimate to each cutoff-specific estimate.

graph_opt(*string*) options to be passed to the graph when **plot** is specified.

Examples

```
Standard use of rdmc
. rdmc yvar xvar, c(cvar)

rdmc with plots of estimates and weights
. rdmc yvar xvar, c(cvar) plot

rdmc showing output from rdrobust and specifying uniform kernel
. rdmc yvar xvar, c(cvar) verbose pooled_opt(kernel(uniform))
```

Saved results

rdmc saves the following in **e()**:

Scalars

<code>e(tau_weight)</code>	weighted estimate
<code>e(se_weight_rb)</code>	robust bias corrected s.e. for weighted estimate
<code>e(pv_weight_rb)</code>	robust bias corrected p-value for weighted estimate
<code>e(ci_weight_l)</code>	left limit of robust bias corrected confidence interval for weighted estimate
<code>e(ci_weight_r)</code>	right limit of robust bias corrected confidence interval for weighted estimate
<code>e(N_h_l)</code>	effective sample size to the left of the cutoff used to estimate weighted estimand
<code>e(N_h_r)</code>	effective sample size to the right of the cutoff used to estimate weighted estimand
<code>e(tau_pool)</code>	pooled estimate
<code>e(se_rb)</code>	robust bias corrected s.e. for pooled estimate
<code>e(pv_rb)</code>	robust bias corrected p-value
<code>e(ci_rb_l)</code>	left limit of robust bias corrected confidence interval
<code>e(ci_rb_r)</code>	right limit of robust bias corrected confidence interval
<code>e(h_l)</code>	bandwidth to the left of the cutoff used to estimate pooled estimand
<code>e(h_r)</code>	bandwidth to the right of the cutoff used to estimate pooled estimand
<code>e(N_h_l)</code>	sample size within bandwidth to the left of the cutoff used to estimate pooled estimand
<code>e(N_h_r)</code>	sample size within bandwidth to the right of the cutoff used to estimate pooled estimand

Matrices

<code>e(b)</code>	bias-corrected coefficient vector
<code>e(V)</code>	robust variance-covariance matrix of the estimators
<code>e(coefs)</code>	conventional coefficient vector
<code>e(pv_rb)</code>	robust p-value vector
<code>e(CI_rb)</code>	bias-corrected confidence intervals
<code>e(H)</code>	vector of bandwidths at each side of each cutoff
<code>e(weights)</code>	vector of weights for each cutoff-specific estimate
<code>e(sampsis)</code>	vector of sample sizes at each side of each cutoff
<code>e(c_failed)</code>	vector of cutoffs at which rdrobust encountered problems

References

- Calonico, S., M. D. Cattaneo, M. H. Farrell, and R. Titiunik. 2017. [rdrubust: Software for Regression Discontinuity Designs](#). *Stata Journal* 17(2): 372-404.
- Calonico, S., M. D. Cattaneo, and R. Titiunik. 2014. [Robust Data-Driven Inference in the Regression-Discontinuity Design](#). *Stata Journal* 14(4): 909-946.
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- Cattaneo, M. D., R. Titiunik, and G. Vazquez-Bare. 2020. [Analysis of Regression Discontinuity Designs with Multiple Cutoffs or Multiple Scores](#). *Stata Journal* 20(4): 866-891.
- Keele, L., and R. Titiunik. 2015. [Geographic Boundaries as Regression Discontinuities](#). *Political Analysis* 23(1): 127-155.

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